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2. Basic Ideas and Terminology
3. Solutions of Differential Equations
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A Simple Population Model

In this section, we consider a simple but important model of population growth whose mathematical formulation leads to separable equations.

This is the **Malthusian growth model**, in which it is assumed that *the rate of increase of the population at any time is proportional to the size of the population at that time*.

Thomas R. Malthus (1766–1834), an English cleric, scholar and economist in the fields of political economy and demography. In 1798 he wrote “An Essay on the Principle of Population”, one of the earliest and most influential books on population.

Let $P(t)$ denote the population at time t. We have

$$\frac{dP}{dt} = kP,$$

where k is a positive constant.

Separating the variables and integrating, we obtain

$$P(t) = P_0 e^{kt},$$

where P_0 is the population at $t = 0$.

This law predicts an exponential increase in the population with time.

This gives a reasonably accurate description of the growth of certain algae, bacteria, and cell cultures.

Definition

1. The formula above is called the Malthusian growth model.

2. The time taken for such a culture to double in size is called the doubling time.

Let's denote by t_d the doubling time. Then $P(t_d) = 2P_0$. This gives

$$2P_0 = P_0 e^{kt_d}.$$

Dividing both sides by P_0 and taking logarithm yields

$$kt_d = \log 2,$$

which implies that the doubling time is

$$t_d = \frac{1}{k} \log 2.$$

Example

The number of bacteria in a certain culture grows at a rate that is proportional to the number present. If the number increased from 500 to 2000 in 2 hours, determine

1. the number present after 12 hours.
2. the doubling time.

Solution. The behavior of the system is formulated by, as above, the differential equation

$$\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt},$$

where the time t is measured in hours.

Taking $t = 0$ as the time when the population was 500, we have $P_0 = 500$. Thus,

$$P(t) = 500e^{kt}.$$

Furthermore, $P(2) = 2000$ which implies that

$$2000 = 500e^{2k},$$

so

$$k = \frac{1}{2}\log 4 = \log 2.$$

Therefore,

$$P(t) = 500e^{t\log 2}.$$

Consequently,

1. The number of bacteria present after 12 hours is

$$P(12) = 500e^{12 \log 2} = 500(2^{12}) = 2,048,000.$$

2. The doubling time of the system is

$$t_d = \frac{1}{k} \log 2 = 1 \text{ hour}.$$

Example

Of the 1,500 passengers, crew, and staff that board a cruise ship, 5 have the flu. After 1 day of sailing, the number of infected people has risen to 10. Assume that the rate at which the flu virus spreads is proportional to the product of the number of infected individuals and the number not yet infected. Determine how many people will have the flu at the end of the 14-day cruise.



Solution.

Let $y(t)$ denote the number of passengers who have the flu at time t.

Then we solve the equation

$$\frac{dy}{dt} = ky(1,500 - y), \quad y(0) = 5, \quad y(1) = 10,$$

where k is a positive constant.

Separating the variables and integrating yields

$$\int \frac{1}{y(1,500 - y)}\, dy = k \int dt.$$

By a partial fraction decomposition of the left hand-side gives

$$\int \left[\frac{1}{1,500y} + \frac{1}{1,500(1,500-y)} \right] dt = k \int dt,$$

so that

$$\frac{1}{1,500} \log \frac{y}{1,500-y} = kt + c,$$

and hence

$$\frac{y}{1,500-y} = c_1 e^{1,500kt}.$$

Imposing the initial condition $y(0) = 5$, we have $c_1 = \frac{1}{299}$.

Then

$$\frac{y}{1,500-y} = \frac{1}{299} e^{1,500kt}.$$

The other initial condition $y(1) = 10$ gives

$$\frac{1}{1,490} = \frac{1}{299} e^{1,500k} \implies k = \frac{1}{1,500} \log \frac{299}{149}.$$

Therefore,

$$\frac{y}{1,500 - y} = \frac{1}{299} e^{t \log(299/149)},$$

from which it follows that

$$y(t) = \frac{1,500 e^{t \log(299/149)}}{299 + e^{t \log(299/149)}} = \frac{1,500}{1 + 299 e^{-t \log(299/149)}}.$$

The final answer then is

$$y(14) = \frac{1,500}{1 + 299 e^{-14 \log(299/149)}} \approx 1,474.$$

It is worth to remark that the Malthusian growth model does not provide an accurate model for the growth of a population over a long time period.

To obtain a more realistic model we need to take into account of the fact that as the population increases, several factor will begin to affect the growth rate.

For example, there will be increased competition for the limited resources that are available, increases in disease, and overcrowding of the limited available space, all of which would serve to slow the growth rate.

In order to model this situation mathematically, we modify the differential equation leading to the simple exponential growth law by adding in a term that slows the growth down as the population increases.

If we consider a closed environment (neglecting factors such as immigration and emigration), then the rate of change of population can be modelled by the differential equation

$$\frac{dP}{dt} = [B(t) - D(t)]P,$$

where $B(t)$ and $D(t)$ denote the birth rate and death rate per individual, respectively.

Note. The simple exponential law corresponds to the case when $B(t) = k$ and $D(t) = 0$.

In the more general situation, the increased competition as the population grows will result in a corresponding increase in the death rate per individual.

Perhaps the simplest way to take account of this is to assume that the death rate per individual is directly proportional to the instantaneous population, and that the birth rate per individual remains constant.

In such a way, we may obtain the so-called **logistic model** - A self-study.

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First-Order Linear Differential Equations

We derive a technique for determining the general solution to any first-order linear differential equation. This is the **most important technique** in the chapter.

Definition

A differential equation that can be written in the form

$$a(x)\frac{dy}{dx} + b(x)y = r(x) \tag{1}$$

where $a(x), b(x)$, and $r(x)$ are functions defined on an interval (a, b), is called a **first-order linear differential equation**.

Assume that $a(x) \neq 0$ on (a, b). We divide both side of (1) by $a(x)$ to obtain the **standard form**

$$\frac{dy}{dx} + p(x)y = q(x), \tag{2}$$

where $p(x) = b(x)/a(x)$ and $q(x) = r(x)/a(x)$.

The idea of the solution technique is to transform the differential equation into the form

$$\frac{d}{dx}[G(x, y)] = F(x)$$

for an appropriate function $G(x, y)$. Then by an integration w.r.t. x gives the general solution to the differential equation.

Example

Solve the differential equation

$$\frac{dy}{dx} + \frac{1}{x}y = e^x, \ x > 0. \tag{3}$$

Solution.

Multiply (3) by x, we have

$$x\frac{dy}{dx} + y = xe^x.$$

But the left-hand side, by the product rule for differentiation, is written as

$$x\frac{dy}{dx} + y = \frac{d}{dx}(xy).$$

So Equation (3) can be rewritten as

$$\frac{d}{dx}(xy) = xe^x.$$

Integrating both sides of this equation w.r.t. x, with the note that here integration by parts is used for the right-hand side $\int xe^x \, dx$:

$$\begin{cases} u = x \\ dv = e^x \, dx \end{cases} \Longrightarrow \begin{cases} du = dx \\ v = e^x, \end{cases}$$

$$\Longrightarrow \int xe^x \, dx = xe^x - \int e^x \, dx = xe^x - e^x + C,$$

we obtain

$$xy = xe^x - e^x + C.$$

Dividing by x (as $x > 0$) gives the general solution to (3)

$$y(x) = e^x - \frac{e^x}{x} + \frac{C}{x} = \frac{1}{x}[e^x(x-1) + C],$$

where C is an arbitrary constant.

Discussion

In the example above, we multiplied the given differential equation by the function $I(x) = x$. It reduced the left-hand side of the resulting differential equation to the integrable form $\frac{d}{dx}(xy)$.

This motivated us to find a nonzero function $I(x)$ such that after multiplying Equation (2) by this function, the left-hand side of the resulting equation

$$I\frac{dy}{dx} + p(x)Iy = Iq(x), \tag{4}$$

can have the form

$$\frac{d}{dx}[I(x)y],$$

here we assume that both functions $p(x)$ and $q(x)$ are continuous on (a, b).

Note that by the product rule,

$$\frac{d}{dx}(Iy) = I\frac{dy}{dx} + \frac{dI}{dx}y. \tag{5}$$

Comparing the left-hand side of Equation (4) and the right-hand side of Equation (5) yields

$$\frac{dI}{dx} = p(x)I. \tag{6}$$

As the last differential equation is separable, we get

$$\log|I| = \int p(x)\,dx + C, \text{ or } I(x) = C_1 e^{\int p(x)\,dx},$$

where C_1 is an arbitrary constant, which can be taken a value 1, since we require only one solution to Equation (6).

Then Equation (2) can be solved, by integrating, as follows

$$\frac{d}{dx}\left[e^{\int p(x)\,dx}y\right] = q(x)e^{\int p(x)\,dx}$$

$$\iff e^{\int p(x)\,dx}y(x) = \left[\int q(x)e^{\int p(x)\,dx}\,dx + C\right]$$

$$\iff y(x) = e^{-\int p(x)\,dx}\left[\int q(x)e^{\int p(x)\,dx}\,dx + C\right].$$

Definition

The function

$$I(x) = e^{\int p(x)\, dx}$$

is called an **integrating factor** for the differential equation (2)

Remark

It is not necessary to memorize the formula said above. In a concrete problem we first evaluate the integrating factor, and then use the resulting equation.

Example

Solve

$$x\frac{dy}{dx} + 2y = \cos x, \ x > 0. \tag{7}$$

Solution

First we write the given differential equation in the standard form, by dividing both sides by x (as $x > 0$)

$$\frac{dy}{dx} + \frac{2}{x}y = \frac{\cos x}{x}.$$

An integrating factor is

$$I(x) = e^{\int \frac{2}{x}\,dx} = e^{2\log x} = x^2.$$

Now multiplying Equation (7) by I, we get

$$\frac{d}{dx}(x^2 y) = x \cos x.$$

Integrating the last equation gives

$$x^2 y = x \sin x + \cos x + C,$$

or

$$y(x) = \frac{1}{x^2}(x \sin x + \cos x + C),$$

where C is an arbitrary constant.

□

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Change of Variables

In this section, we consider two further types of differential equations which can be solved by using a **change of variables** to **reduce** them to one of the types we know how to solve.

Definition

A function $f(x, y)$ is said to be **homogeneous of degree zero** if

$$f(tx, ty) = f(x, y)$$

for all positive values of t for which (tx, ty) is in the domain of f.

Equivalently, f is homogeneous of degree zero if it is invariant under a re-scaling of the variables x and y.

Examples

1. The simplest non-constant functions which are homogeneous of degree zero are $f(x,y) = \dfrac{y}{x}$ and $f(x,y) = \dfrac{x}{y}$.

   Indeed, for say $f(x,y) = \dfrac{y}{x}$ we have

   $$f(tx,ty) = \frac{ty}{tx} = \frac{y}{x} = f(x,y).$$

2. If $f(x,y) = \dfrac{x^2 - y^2}{2xy + y^2}$, then

   $$f(tx,ty) = \frac{t^2(x^2 - y^2)}{t^2(2xy + y^2)} = f(x,y),$$

   so that f is homogeneous of degree zero.

Note that in the second example, dividing both the numerator and denominator by x^2, we have

$$f(x,y) = \frac{x^2 - y^2}{2xy + y^2} = \frac{\frac{x^2}{x^2} - \frac{y^2}{x^2}}{\frac{2xy}{x^2} + \frac{y^2}{x^2}} = \frac{1 - \left(\frac{y}{x}\right)^2}{2\left(\frac{y}{x}\right) + \left(\frac{y}{x}\right)^2}.$$

This means that f can be considered to depend on the single variable $V = \frac{y}{x}$.

Actually, we have the following result that establishes the basic property of all functions that are homogeneous of degree zero.

Theorem

A function $f(x, y)$ is homogeneous of degree zero if and only if it depends on either $\frac{y}{x}$ or $\frac{x}{y}$ only.

Homogeneous first-order differential equations

We now consider differential equations that satisfy the following definition.

Definition

If $f(x, y)$ is homogeneous of degree zero, then the differential equation

$$\frac{dy}{dx} = f(x, y) \tag{1}$$

is called a **homogeneous first-order differential equation**.

In general, a homogeneous first-order differential equation cannot be solved directly.

However, we can solve it, by the theorem above, using a change of variables: instead of using variables x and y, we can use the variables x and V, where $V = \dfrac{y}{x}$, that is $y = xV(x)$.

Theorem

The change of variables $y = xV(x)$ reduces a homogeneous first-order differential equation $\dfrac{dy}{dx} = f(x, y)$ to the separable equation

$$\frac{1}{F(V) - V} dV = \frac{1}{x} dx,$$

where F is an appropriate function.

Proof

Exrecise.

Remark

The separable equation said above can be integrated to obtain a relationship between V and x. Then the solution to the given differential equation can be obtained by substituting $\frac{y}{x}$ for V in this relationship.

Examples

Find the general solution to

$$\frac{dy}{dx} = \frac{4x + y}{x - 4y}.$$

Solution

Since $f(x, y) = \frac{4x+y}{x-4y}$ is homogeneous of degree zero, the given equation is a first-order homogeneous differential equation.

Substituting $y = xV$ into the equation yields

$$\frac{d}{dx}(xV) = \frac{4+V}{1-4V},$$

that is

$$x\frac{dV}{dx} + V = \frac{4+V}{1-4V},$$

or equivalently,

$$x\frac{dV}{dx} = \frac{4(1+V^2)}{1-4V}.$$

Separating the variables gives

$$\frac{1-4V}{4(1+V^2)}dV = \frac{1}{x}dx.$$

We write this as

$$\left[\frac{1}{4(1+V^2)} - \frac{V}{1+V^2}\right] dV = \frac{1}{x} dx.$$

Now we integrate both side directly. For this the following integral formulas are used:

(1) $\int \frac{dx}{1+x^2} = \arctan x + C$, and (2) $\int \frac{x\, dx}{1+x^2} = \frac{1}{2}\log(1+x^2) + C.$

The first integral is standard: the change of variables $x = \tan t$ works well.

For the second integral: putting $1 + x^2 = t$ we have $dt = 2x\, dx$, and hence

$$\int \frac{x\, dx}{1+x^2} = \frac{1}{2}\int \frac{dt}{t} = \frac{1}{2}\log|t| + C = \frac{1}{2}\log(1+x^2) + C.$$

Thus we obtain

$$\frac{1}{4}\arctan V - \frac{1}{2}\log(1+V^2) = \log|x| + C.$$

Substituting back $V = \dfrac{y}{x}$ and multiplying through by 2 yields

$$\frac{1}{2}\arctan\left(\frac{y}{x}\right) - \log\left(\frac{x^2+y^2}{x^2}\right) = \log(x^2) + C_1,$$

where $C_1 = 2C$.
Moving $\log\left(\frac{x^2+y^2}{x^2}\right)$ to the right-hand side yields the answer

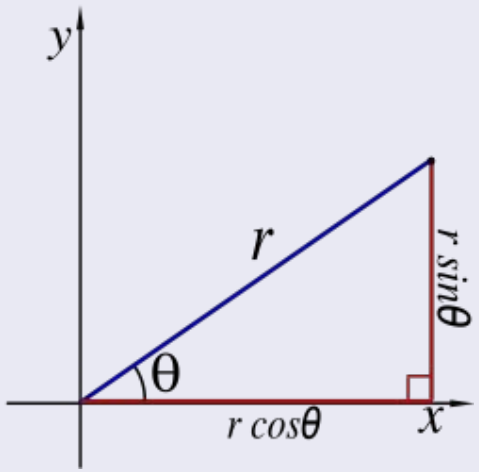
$$\frac{1}{2}\arctan\left(\frac{y}{x}\right) - \log(x^2+y^2) = C,$$

where C is an arbitrary constant.

□

Remark

The solution to the example above is more easily expressed in terms of polar coordinates.



Indeed, note that $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases} \iff \begin{cases} r = \sqrt{x^2+y^2} \\ \theta = \arctan\left(\frac{y}{x}\right), \end{cases}$ we then obtain a logarithmic spiral:

$$r = Ce^{\frac{\theta}{4}}$$

Indeed, substituting into the equation above yields

$$\frac{1}{2}\theta - \log(r^2) = C_1,$$

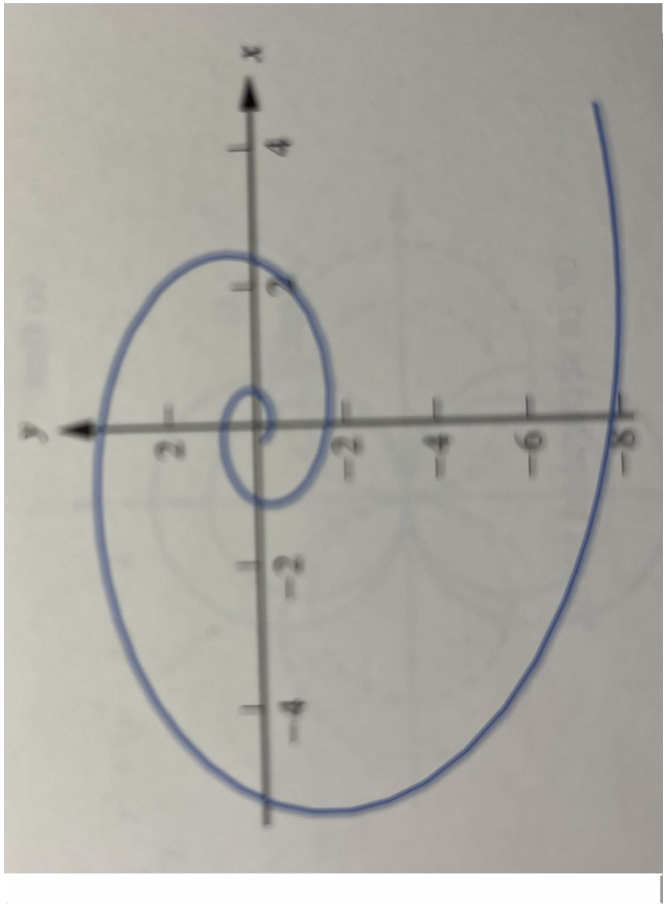
or equivalently,

$$\log r = \frac{1}{4}\theta + C_2.$$

Exponentiating both sides of the equation gives

$$r = Ce^{\frac{\theta}{4}}.$$

For each value of C, this is the equation of a logarithmic spiral. The particular spiral with equation $r = \frac{1}{2}e^{\frac{\theta}{4}}$ is shown below.


y
x
4
2
-2
-4
-6
-8
-2
-4

Bernoulli equations

We now consider a special type of **nonlinear** differential equation that can be reduced to a **linear** equation by a change of variables.

Definition

A differential equation that can be written in the form

$$\frac{dy}{dx} + p(x)y = q(x)y^{\alpha}, \tag{2}$$

where α is a ***real*** constant, is called a **Bernoulli equation**.

If $\alpha = 0$ or $\alpha = 1$, Equation (2) is linear, but otherwise it is nonlinear.

In general, we can reduce Equation (2) to a linear equation as follows.

- Divide Equation (2) by y^{α} to obtain

$$y^{-\alpha}\frac{dy}{dx} + y^{1-\alpha}p(x) = q(x). \tag{3}$$

- Make the change of variables $u(x) = y^{1-\alpha}$ to get

$$\frac{du}{dx} = (1-\alpha)y^{-\alpha}\frac{dy}{dx}.$$

Thus

$$y^{-\alpha}\frac{dy}{dx} = \frac{1}{1-\alpha}\frac{du}{dx}.$$

Now substituting into Equation (3) the expressions $y^{1-\alpha} = u(x)$ and $y^{-\alpha}\dfrac{dy}{dx} = \dfrac{1}{1-\alpha}\dfrac{du}{dx}$, we have the **linear** differential equation

$$\frac{1}{1-\alpha}\frac{du}{dx} + p(x)u = q(x),$$

or in the **standard** form

$$\frac{du}{dx} + (1-\alpha)p(x)u = (1-\alpha)q(x).$$

Examples

Solve

$$\frac{dy}{dx} + \frac{3}{x}y = \frac{12y^{\frac{2}{3}}}{\sqrt{1+x^2}},\ x > 0.$$

Solution

The differential equation is Bernoulli equation with $\alpha = \frac{2}{3}$.
Dividing both sides of the equation by $y^{\alpha} = y^{\frac{2}{3}}$ we obtain

$$y^{-\frac{2}{3}}\frac{dy}{dx} + \frac{3}{x}y^{\frac{1}{3}} = \frac{12}{\sqrt{1+x^2}}. \tag{4}$$

We make the change of variables

$$u = y^{1-\alpha} = y^{\frac{1}{3}}.$$

Then we get

$$\frac{du}{dx} = \frac{1}{3} y^{-\frac{2}{3}} \frac{dy}{dx}.$$

Now substitute the last two expressions into Equation (4) yields

$$3\frac{du}{dx} + \frac{3}{x}u = \frac{12}{\sqrt{1+x^2}},$$

or in the standard form

$$\frac{du}{dx} + \frac{1}{x}u = \frac{4}{\sqrt{1+x^2}}. \tag{5}$$

This is a **first-order linear** differential equation, whose integrating factor is

$$I(x) = e^{\int \frac{1}{x} dx} = e^{\log x} \text{ (because } x > 0) \ = x.$$

So Equation (5) can be written as

$$\frac{d}{dx}(xu) = \frac{4x}{\sqrt{1+x^2}}.$$

Integrating this equation, we obtain

$$xu(x) = 4\sqrt{1+x^2} + C.$$

Here we use the integral formula

$$\int \frac{x \, dx}{\sqrt{1+x^2}} = \sqrt{1+x^2} + C,$$

which is computed as follows: putting $1 + x^2 = t$ *implies* $x \, dx = \frac{dt}{2}$;

then

$$\int \frac{x\,dx}{\sqrt{1+x^2}} = \int \frac{dt}{2\sqrt{t}} = \sqrt{t} + C = \sqrt{1+x^2} + C.$$

Returning back to the obtained result

$$xu(x) = 4\sqrt{1+x^2} + C,$$

we have

$$u(x) = \frac{1}{x}\left(4\sqrt{1+x^2} + C\right),$$

and the solution to the original differential equation is

$$y^{\frac{1}{3}} = \frac{1}{x}\left(4\sqrt{1+x^2} + C\right),$$

where C is an arbitrary constant.

□